UDROŻNIENIE ŁÓDZKIEGO WĘZŁA KOLEJOWEGO (TEN-T), ETAP II, ODCINEK ŁÓDŹ FABRYCZNA - ŁÓDŹ KALISKA/ŁÓDŹ ŻABIENIEC – PROJEKT POIIŚ 5.1-15

TBM 13.08 M - CONSTRUCTION FOLLOW-UP

IDENTIFICATION TABLE	
Client	PBDiM - Energopol
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AUTHOR		REV
Opracowal	Andrea Luciani	00
Projektant	Paolo Cucino - Inżynieryjna mostowa i konstrukcyjno-budowlana, upr. nr ŁOD/TR/0247/19	00 SWS Engineering S.p.A. Via della Stazione, 27 Fraz. Mattarello - 38123 TRENTO Cod. Fisc. e P.IVA: 00580910222

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1. INTRODUCTION

S-1222 TBM EPB machine with 13.08 m excavation diameter is currently mining Axis 23, in the section between Polesie Station and Śródmieście Station. TBM has restarted from chainage 1+962.

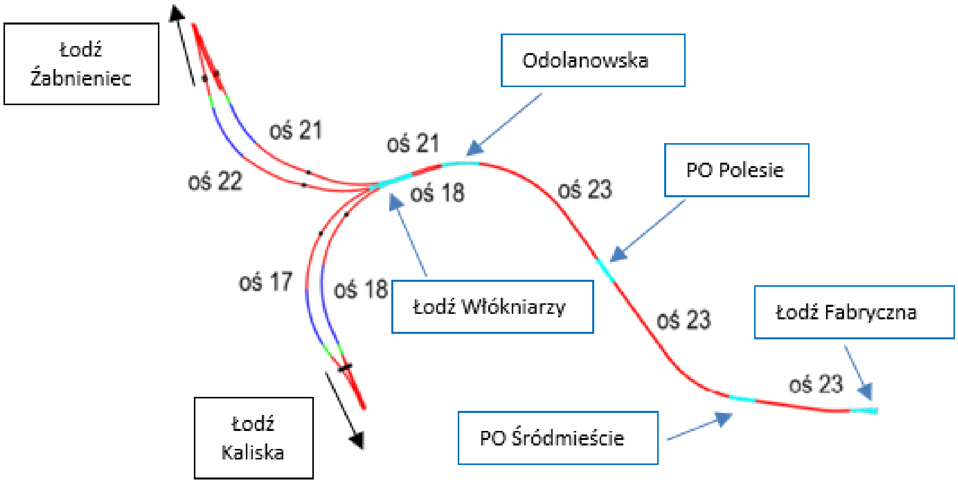

Figure 1-1 Overview of main structural elements of the Project

This daily follow-up report aims to give a general overview of S-1222 TBM drive during 26/08/2024. It includes EPB face pressure, belt scale parameters, thrust and torque values, primary grout parameters, guidance, as well as monitoring follow-up data.

According to VMT system, TBM is at Chainage 1+706.52 after the erection of ring n. 220; which corresponds to a total excavation length of 259.95 m from the starting chainage.

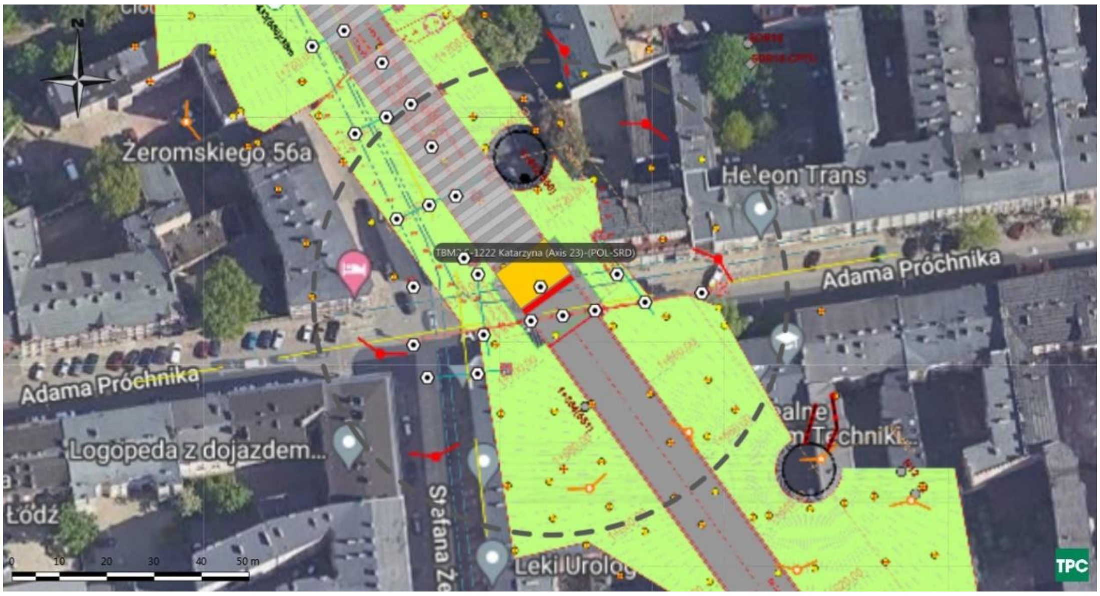

Figure 1-2 Plan view of TBM Position on 27/08/2024

2. FACE PRESSURES AND EXCAVATION VOLUMES

Average EPB pressure at the crown is at about 1.8 bar. The value is within the attention values. There have been a rapid drop of face pressure during ring excavation 218 due to blockage of the screw conveyor gate.

Figure 2-1 shows EPB sensors.

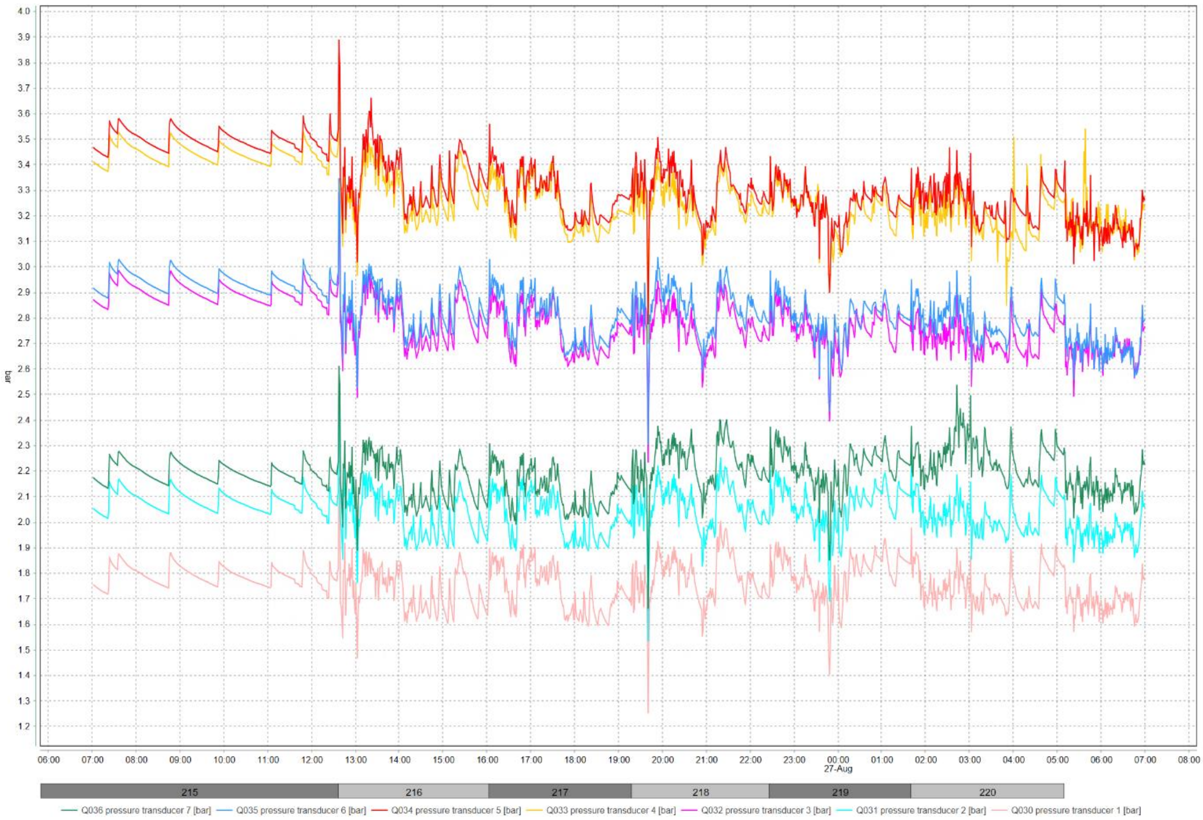

Figure 2-1 EPB pressure sensor

The delta between pressure transducer 1 and 2 has an average value of 0.28 bar indicating that the bulk chamber is correctly filled with material.

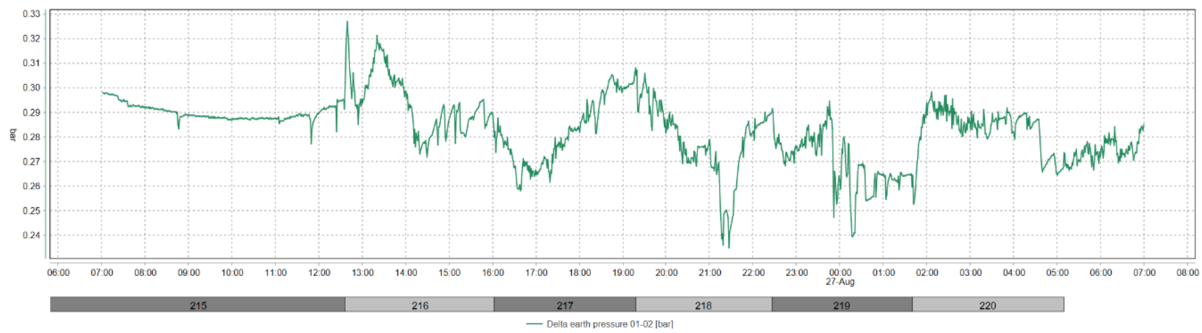

Figure 2-2 Delta between pressure sensors 1 and 2

Belt scale n. 2 show values much lower than n. 1. The results of Belt scale n. 2 seems to be affected by a constant error and are therefore considered not representative. The belt scale value is a raw indication of the excavated material quantity.

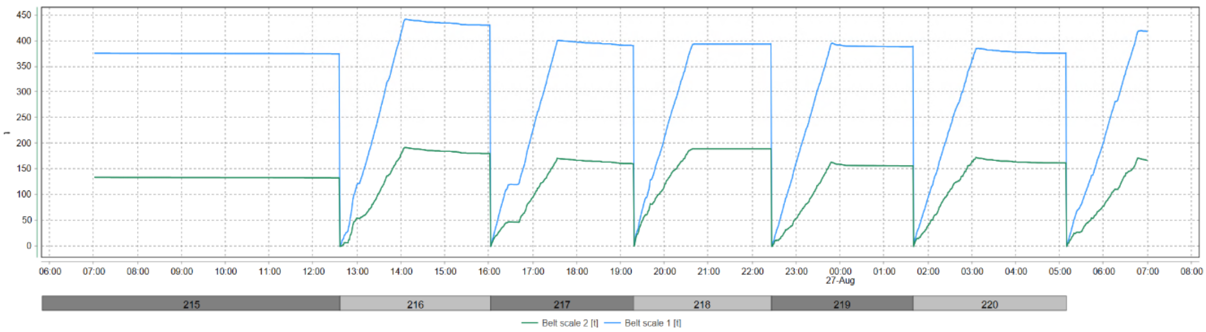

Figure 2-3 Belt scales

Table 2-1 Belt scale values at the end of each advance

Ring	Belt scale 1 (t)	Belt scale 2 (t)
215	377	135
216	443	193
217	401	171
218	395	190
219	396	164
220	386	173

3. THRUST AND CUTTERHEAD TORQUE

During excavation, the average torque is of 4.5 MNm, but is increasing to 7 MNm.

During advancement, average advance rate is of 17 mm/min.

During excavation, thrust force shows an average value of 62.6 MN and a maximum of 77.7 MN.

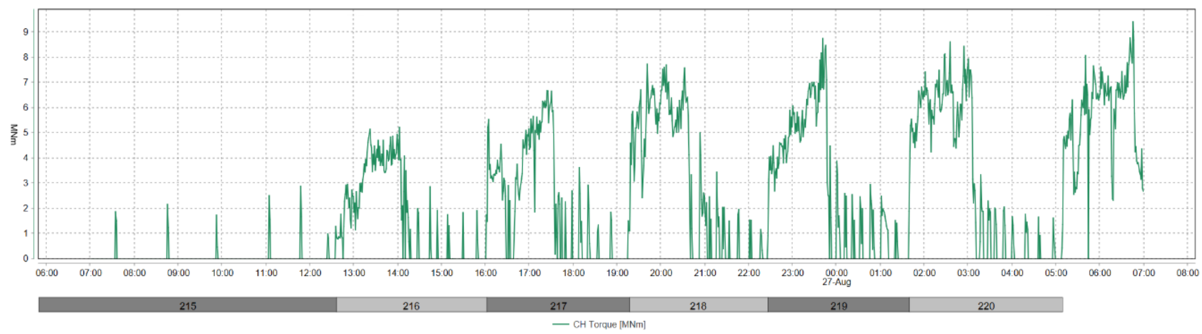

Figure 3-1 CH Torque

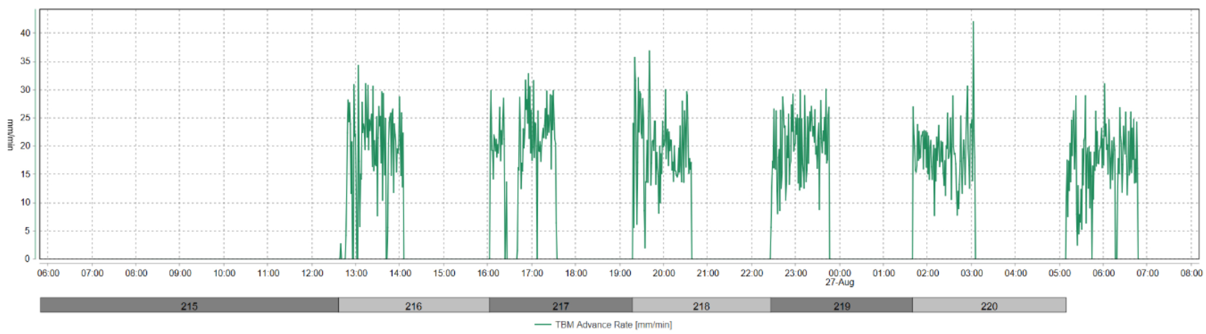

Figure 3-2 Advance rate

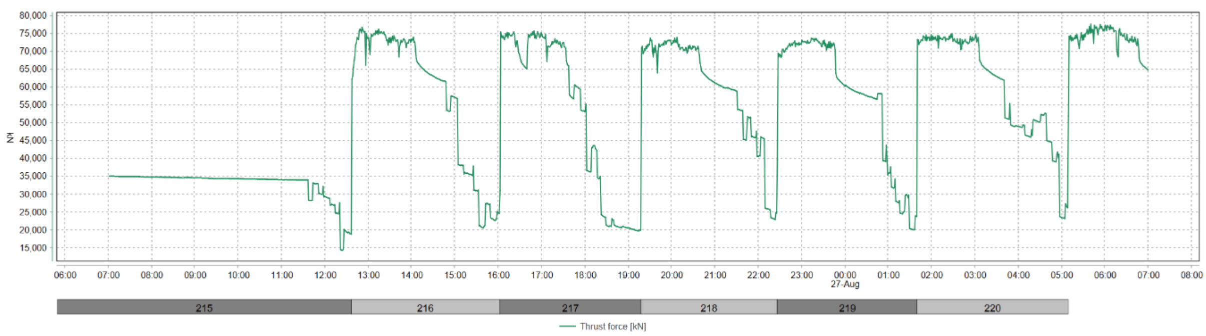

Figure 3-3 Thrust force

4. GROUT INJECTION

Grout injection has a volume above the minimum theoretical annular volume +%10 of 13.54 m^3. Ration B/A is at 7%.

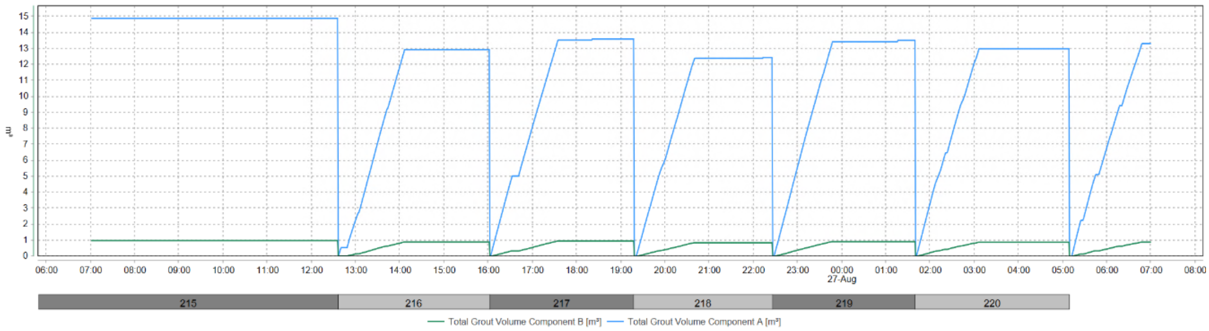

Figure 4-1 Grout volumes

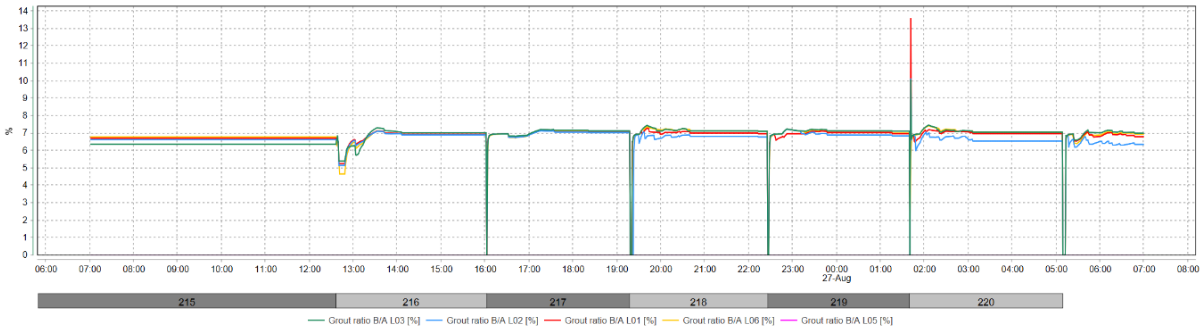

Figure 4-2 Volume ratio Components B/A

Table 4-1 Grout volumes per ring

Ring	Component A (m^3)	Component B (m^3)	B/A (%)
215	14.9	1	7
216	13	0.9	7
217	13.6	1	7
218	12.5	0.9	7
219	13.5	0.9	7
220	13	0.9	7

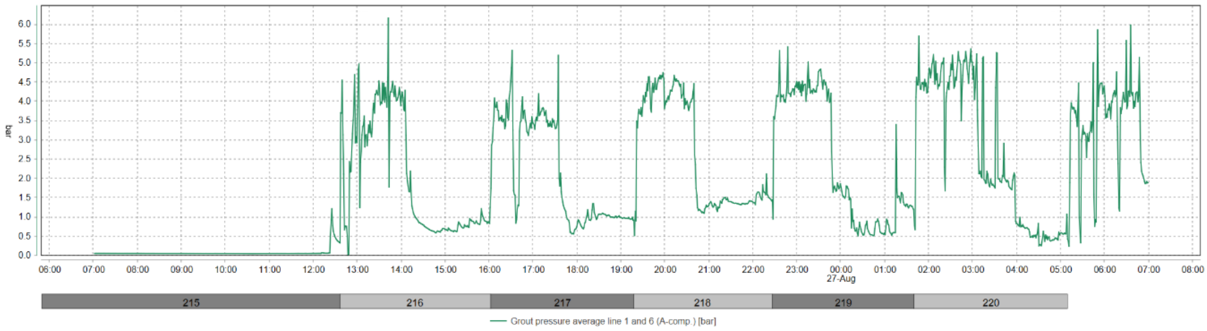

Figure 4-3 Average injection pressure line 1 and 6

5. GUIDANCE

The TBM vertical deviation is of about 6.11 mm on the front edge of the shield and the horizontal deviation of about 11.49 mm (Figure 5-1). Maximum radial deviation is of 21.11 mm.

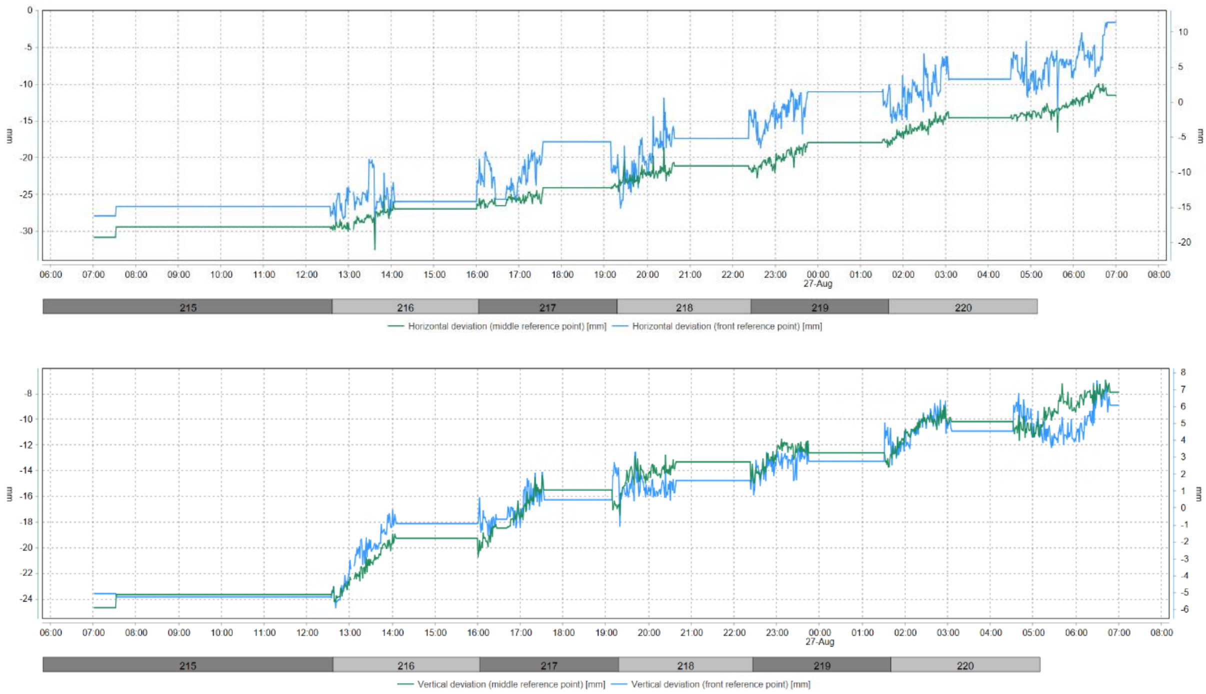

Figure 5-1 Machine deviation from the alignment

6. MONITORING

According to Geoscope, settlements of the buildings above the tunnel are kept within the limits.

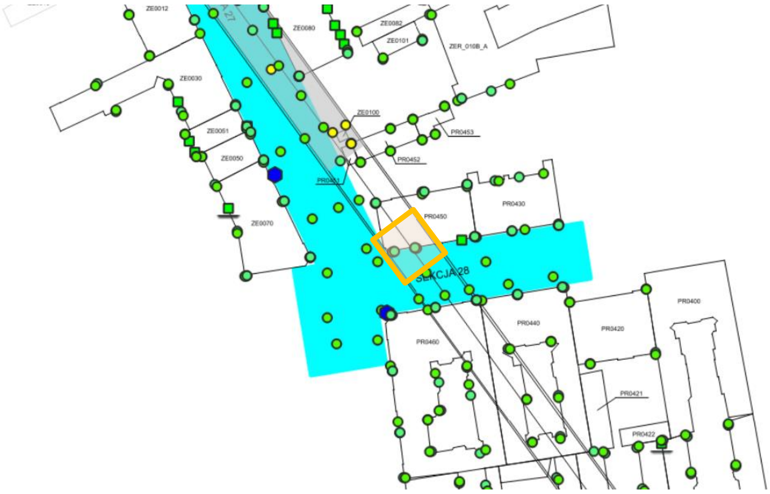

Figure 6-1 Monitoring points

The monitoring points on Zeromskiego street are stable.

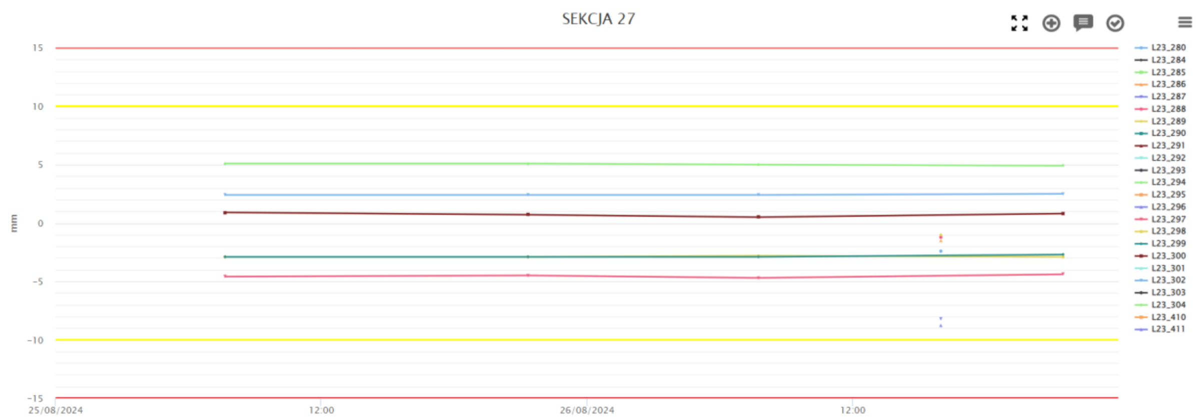

Figure 6-2 Settlement of Section on Zeromskiego street

The monitoring points on Prochnika street are stable.

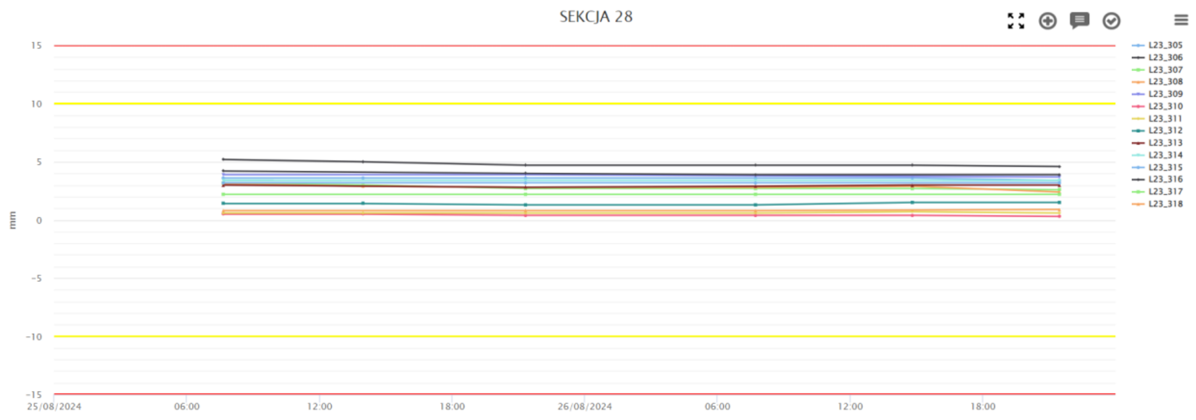

Figure 6-3 Settlement of Section on Prochnika street

Building ZE0050 is stable.

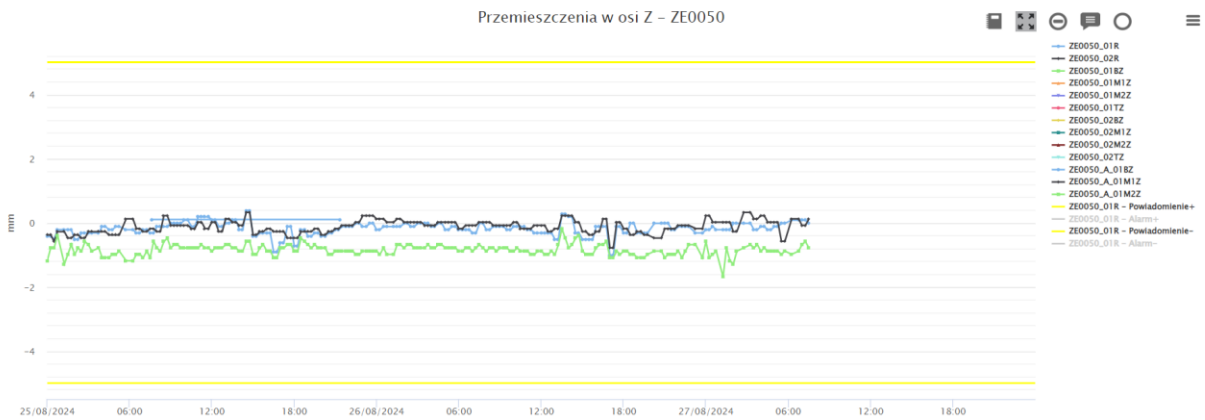

Figure 6-4 Settlement of building ZE0050

In buildings ZE0100, monitoring point are stable.

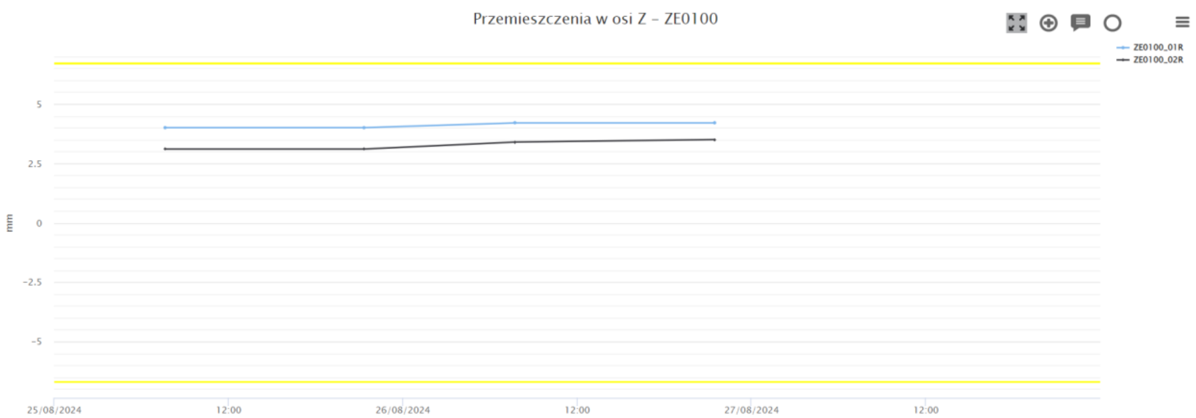

Figure 6-5 Settlement of building ZE0100

In building PR0452, moitoring points are stable.

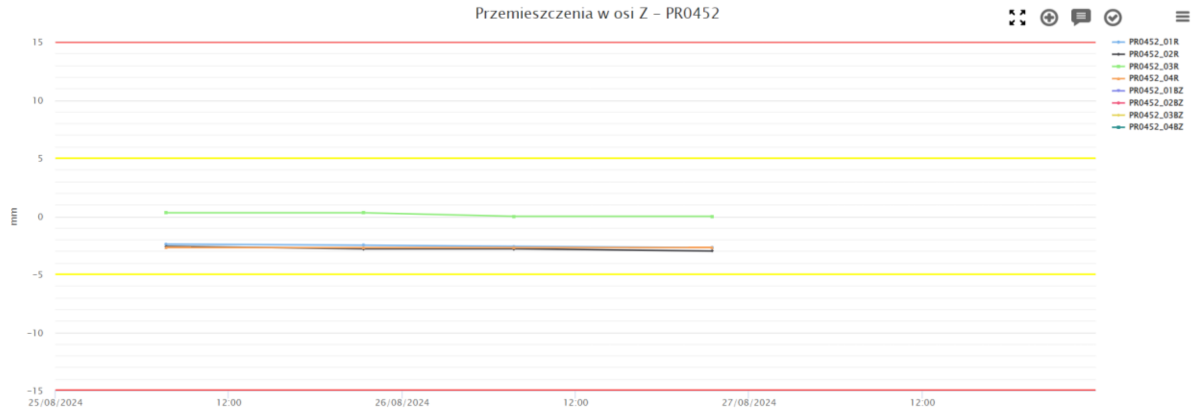

Figure 6-6 Settlement of building PR0452

On PR0450, monitoring points has settled of about 1.5 mm in the last 24 hours.

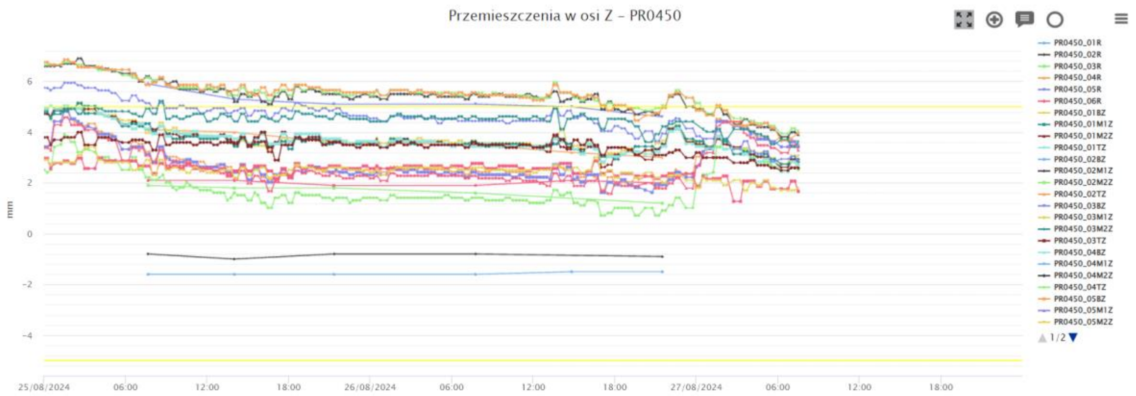

Figure 6-7 Settlement of building PR0450

On ZE0070, monitoring points values are stable.

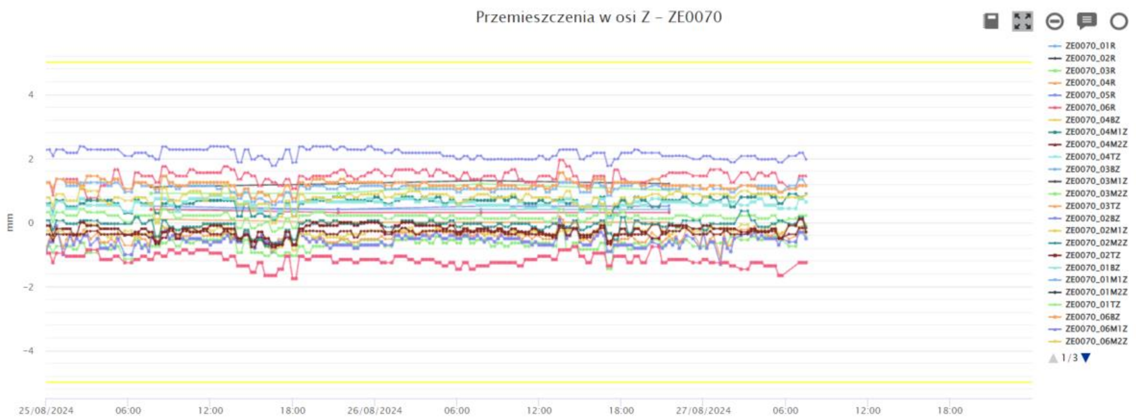

Figure 6-8 Settlement of building ZE0070

On PR0430, monitoring points are stable.

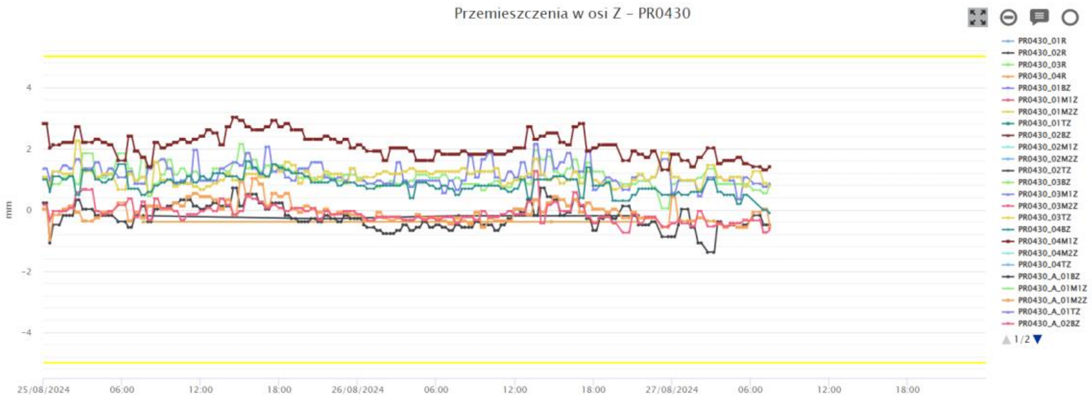

Figure 6-9 Settlement of building PR0430

On PR0460, monitoring points show a small downward trend.

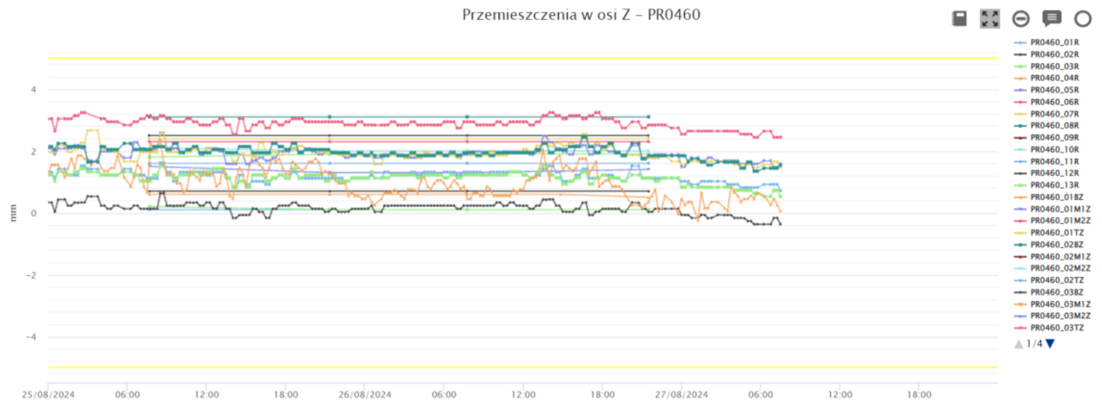

Figure 6-10 Settlement of building PR0460

On PR0460, monitoring points are stable.

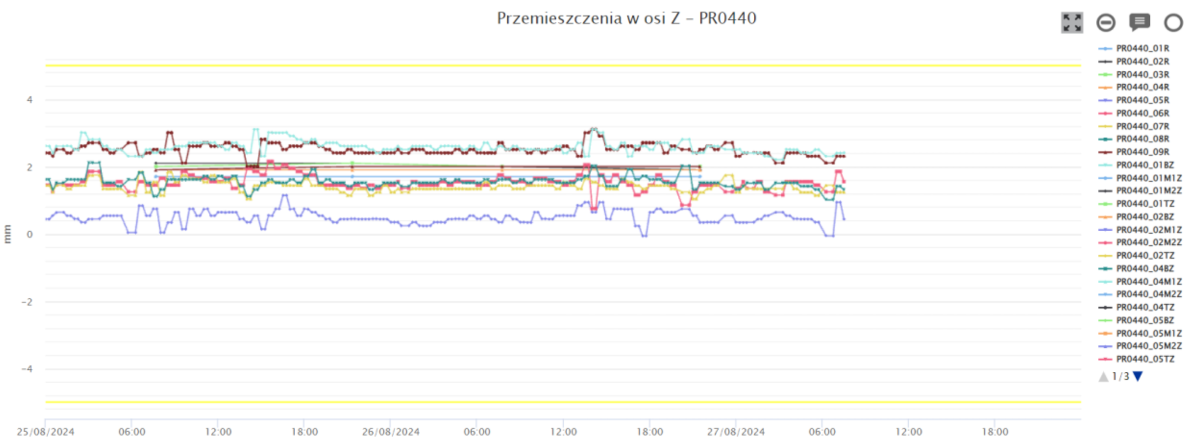

Figure 6-11 Settlement of building PR0440

7. COMPENSATION GROUTING ACTIVITIES

According to Raport Dobowy 26/08/2024 of Soletanche, compensation grouting has been performed on the 26/08/2024 from Shaft 2 in Próchnika 45.

The total amount of declared injected volume is 2736 l.

8. SOIL CONDITIONING

The volume of injected foam is plotted in the following Figure. The FIR value is computed based on the excavated material amount per ring and reported in the Table.

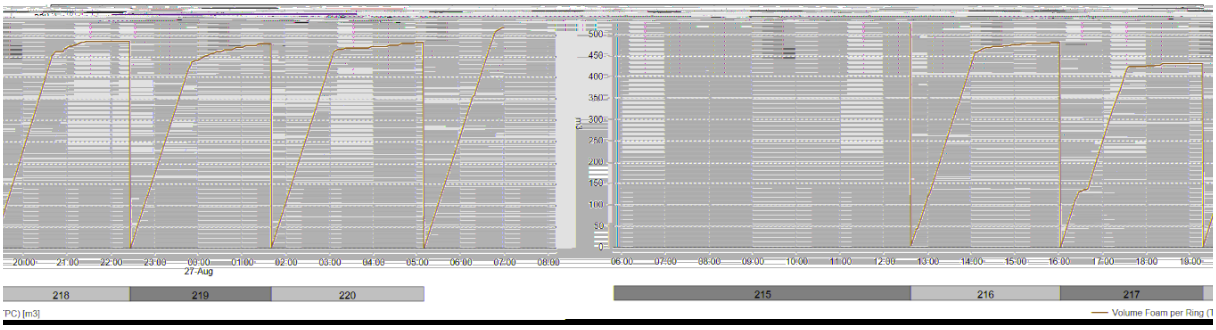

Figure 8-1 Injected foam volume

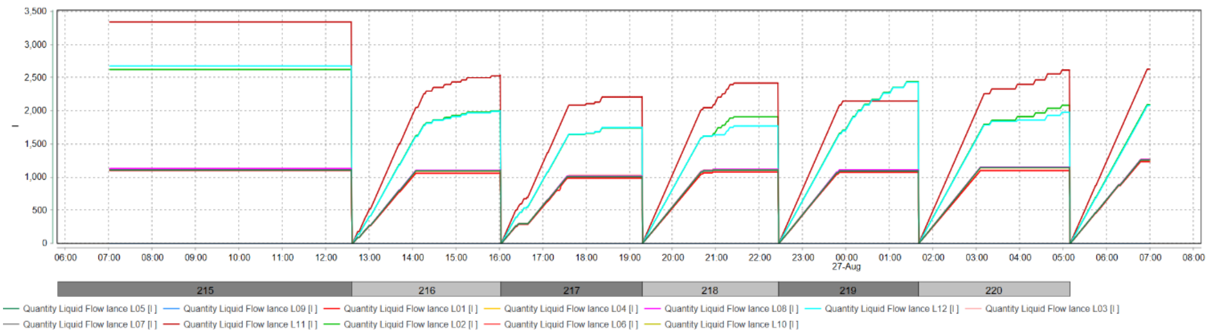

Figure 8-2 Conditioning liquid volume

Table 8-1 Soil conditioning data

Ring	Liquid volume (m^3)	C (%)	FIR (%)
215	17.6	1.9	254
216	15.4	2	227
217	13.9	2	204
218	15	2	229
219	15.9	2	227
220	15.8	2	228